

TIC 111240209

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

BLS EPHEMERIS

Period 33.245106 d (ALIASED, see below)
-> corrected period: 66.490212 d
Epoch (t0) 2240.2131 BTJD
Duration 0.2440 d
Depth 16.214 % (162,137 ppm)
SNR 1156.6

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

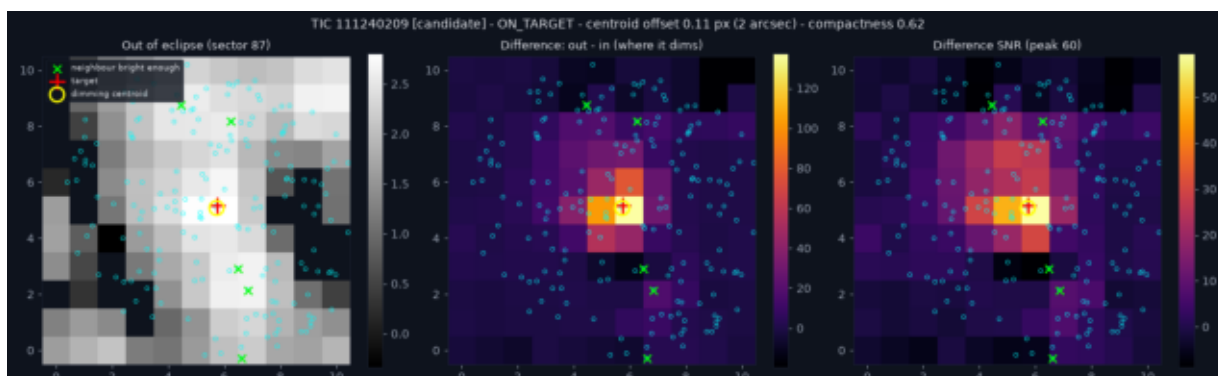
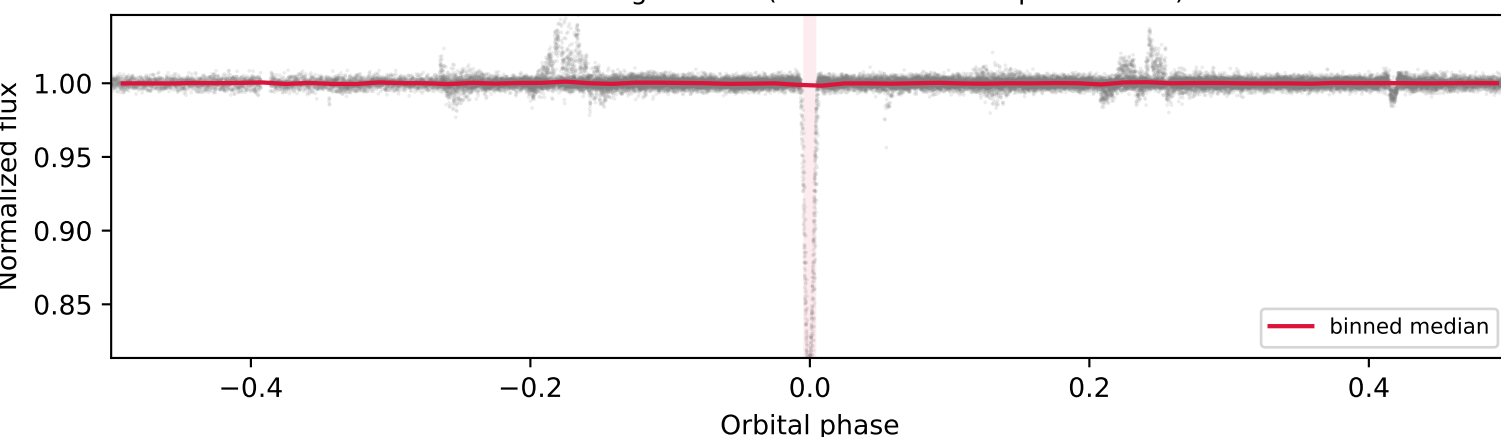
Eclipses observed 4 (gate ≥ 3)
Robust depth ratio 1.06 (gate ≥ 0.50)
Robust depth sigma 903.5 (gate ≥ 7.0)
Odd/even z-score 39.49 (ALIAS FLAGGED)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 87 light-curve dip in this sector: 101.3 sigma t0 refined by -0.0025 phase
Difference-image peak SNR 59.53 (gate ≥ 5.0) Compactness 0.62 (gate ≥ 0.50)
Dimming centroid offset 0.11 px (2 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 111240236
2.29 px away, dTmag +1.10 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 39.49 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 66.490212 d, in any write-up.